

On the Decoder Bottleneck in Physics-Based Emotion-Conditioned Motion Generation

Final Project Report – EmoMoconvq

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Abstract—We investigate whether MoConVQ [6]’s frozen physics-based text-to-motion model can be retrofitted with discrete emotion control via parameter-efficient fine-tuning. We propose EmoMoconvq, which adds a small emotion embedding to T2M-MoConGPT’s text features and distils emotion knowledge from the baseline model’s responses to text containing emotion adverbs. Through three progressive experimental variants we obtain a clear, reproducible result: the discrete emotion label *does* produce statistically detectable variation in the token output of the fine-tuned GPT (separation ratio $1.36\times$ baseline noise; 39.3% 3-class classifier accuracy, with happy reaching 50% individually). However, when those tokens are decoded through MoConVQ’s frozen physics decoder, six standard affective kinematic features (vertical energy, step frequency, lean angle, etc.) show *no* significant variation across emotions (one-way ANOVA $p > 0.7$ for all features; paired analysis $p > 0.26$). We diagnose this as a *decoder bottleneck*: the physics tracker, optimised for kinematic fidelity to a single training distribution, normalises emotion-induced token differences out of the physical motion. We support the diagnosis with qualitative video output, release a working Gradio demo for interactive emotion-conditioned generation, and discuss what would be required to unlock real emotion-expressive physical motion in this class of model.

I. INTRODUCTION

Physics-based motion generators are increasingly competitive with kinematic-only models because they guarantee dynamic plausibility: the generated motion is executed by an articulated character under contact and gravity. The current state of the art, MoConVQ [6], generates physically simulated motion from text via a residual-VQ representation combined with a GPT-style transformer (T2M-MoConGPT) and a frozen physics tracking policy.

This expressive power comes with a control limitation: the model takes unstructured natural-language text as its only conditioning signal. A downstream system that wants to “set the emotional state” of an animated character must phrase that state in text. This is brittle (the exact wording matters; many emotion-action pairs are out-of-distribution) and it conflates emotion with all other semantic content the text carries.

Our question. Can we add a small *discrete* emotion interface to MoConVQ’s existing text-to-motion stack, such that the same text prompt can be rendered in different emotional styles by setting an explicit `emotion` flag?

Our approach. We freeze the entire pretrained MoConVQ model and attach a small emotion embedding module ($\sim 0.8M$ trainable parameters) that perturbs the T5 text feature before it enters the temporal cross-attention layers (§III). Because no public dataset of emotion-labelled physical motion exists in MoConVQ’s character format, we adopt a *distillation* strategy:

we run the baseline T2M-MoConGPT on text that already contains explicit emotion adverbs (*walks happily*, *walks sadly*...) to produce token sequences, then train EmoMoconvq to reproduce those same token sequences from the *stripped* text plus the discrete emotion label (§III-C). Through three progressive variants we tighten the experimental design until the emotion label is provably the only useful conditioning signal in training (§III-D).

Our finding. Token-level evaluation (§IV-B) confirms that the emotion module *does* learn to route information through the discrete label: on a held-out compositional test set of HumanML3D [1] captions containing no emotion words, EmoMoconvq’s outputs exhibit emotion-induced variation $1.36\times$ stronger than baseline stochasticity, and a trained 3-class classifier picks up the conditioned emotion at 39.3% ($> 33.3\%$ chance), with the *happy* class reaching 50%. However, kinematic ANOVA on the physically-decoded motions (§IV-C) finds *no* significant emotion effect on six standard affective features. We diagnose this as the **physics-decoder bottleneck** (§V): the frozen tracking policy collapses the emotion-induced latent variance back to its training distribution, washing out the very signal our module produces.

Contributions. (i) A clean, non-invasive emotion-conditioning architecture for MoConVQ that is a strict parameter superset of the pretrained checkpoint. (ii) A reproducible three-stage distillation methodology that exposes the emotion module’s true capability without confounding text-memorisation. (iii) A token-level vs. motion-level decomposition of the evaluation that isolates the decoder as the bottleneck. (iv) Negative result with mechanistic diagnosis pointing to concrete future work (decoder fine-tuning, motion-level supervision). (v) A working Gradio interactive demo and 15-clip qualitative comparison video set.

II. RELATED WORK

Physics-based motion generation. MoConVQ [6] encodes motion into residual VQ tokens and decodes them via a model-based RL policy that drives an ODE physics simulator, trained on 23 hours of motion data. Its frozen encoder and codebook make it well-suited for lightweight downstream adaptation, which we exploit.

Text-to-motion generation. T2M-GPT [7] and MDM [5] generate kinematic motion from text using discrete transformers and diffusion respectively. Both are trained on HumanML3D [1], are not physics-based, and do not expose an explicit emotion control.

Style and emotion conditioning. ACTOR [3] conditions a transformer VAE on action labels; classical motion-style-transfer methods (Holden et al.; Aberman et al.) decompose content and style from style-labelled mocap. Our work differs in being non-invasive (frozen backbone) and in using *text-prompt distillation* in lieu of style-labelled motion.

Parameter-efficient adaptation. We draw on the LoRA/adaptor family [2] in spirit. Our emotion module is even lighter: $\sim 0.8\text{M}$ parameters with the rest of the model frozen. Path B of our experiments (§III-D) is essentially a $\sim 0.4\%$ adapter.

III. METHOD

A. System overview

MoConVQ’s text-to-motion pipeline has three frozen components: (1) a 1D convolutional motion encoder + 8-layer residual VQ codebook ($N = 512$, $d = 768$); (2) T2M-MoConGPT, a dual transformer with a 12-layer temporal block and a 5-layer depth block predicting RVQ indices; and (3) a deconvolutional decoder + mixture-of-experts tracking policy + ODE physics simulator. We touch only the temporal transformer’s input (its T5 text feature) plus, optionally, its top few layers.

B. Emotion embedding module

Let $\mathcal{C} = \{\text{neutral, happy, sad, angry, fearful}\}$ be the 5-class emotion set. We learn an embedding table $E_e : \mathcal{C} \rightarrow \mathbb{R}^{d_e}$ ($d_e = 512$) and a two-layer MLP projection $\text{MLP}_{\text{proj}} : \mathbb{R}^{d_e} \rightarrow \mathbb{R}^{d_{\text{T5}}}$ that lifts the emotion vector to the 1024-dimensional T5-large [4] feature space consumed by cross-attention. The conditioning signal seen by every temporal block is

$$f_{\text{cond}} = f_{\text{T5}} + \text{MLP}_{\text{proj}}(E_e(c)). \quad (1)$$

The last linear of MLP_{proj} is zero-initialised so that $f_{\text{cond}} = f_{\text{T5}}$ at step 0 and the model degrades gracefully to the pretrained baseline. The whole module is implemented as a subclass of MoConVQ’s transformer; the upstream source code is unmodified. Loading the pretrained checkpoint into the subclass with `strict=False` leaves exactly the 5 emotion tensors uninitialised and produces no unexpected keys.

C. Distillation training objective

A complete public dataset of (text, emotion, MoConVQ-tokens) triples does not exist (§IV-A). We instead distil from the baseline:

- 1) Pick a set of *base* captions b that contain no emotion words.
- 2) For each base b and each emotion $e \in \mathcal{C}$, form the *augmented* caption $b^{(e)}$ by appending the canonical emotion adverb (*walks* $\rightarrow$ *walks happily*).
- 3) Run the frozen baseline T2M-MoConGPT on each $b^{(e)}$ to obtain a target token sequence $I^{(b,e)}$ and the corresponding quantised latent sequence.
- 4) Train EmoMoconvq to predict $I^{(b,e)}$ given (b, e) – crucially, the *stripped* base text plus the emotion label, never the augmented text.

The training objective is the same per-token NLL loss as MoConGPT, implemented from scratch (MoConVQ does not release training code).

Because b is the same across all five emotions but the targets $I^{(b,e)}$ differ, the text input alone cannot predict the target: the model is *forced* to use the discrete emotion label.

D. Three progressive experimental variants

We ran three increasingly tight variants to disentangle “the module learnt something” from “the module is the only thing that could have learnt it”.

v1 – uncontrolled distillation, top-4 layers. 463 unique emotion-bearing captions filtered from HumanML3D, one emotion per caption. The model can in principle memorise the text $\rightarrow$ tokens map purely from text features and ignore the emotion label. 47.3M trainable parameters (24.3% of the model).

v2 – 5-variant distillation, top-2 layers. 100 base captions $\times$ 5 emotions = 500 samples. Same base text, five different targets, so the emotion label is the only distinguishing input. Train/val split is *base-stratified* so held-out captions are never seen in training. 24.0M trainable (12.4%).

Path B – emotion-module only. v2 dataset, but every temporal layer is also frozen; only E_e and MLP_{proj} are trainable. 0.79M trainable parameters (0.41%).

3-class D’. v2 with the most challenging classes (*sad, angry*) removed – the proposal’s pre-registered fallback. 300 samples on {neutral, happy, fearful}; chance level 33.3% (vs. 20%).

IV. EXPERIMENTS

A. Datasets

Why distillation in the first place. The original proposal called for HumanML3D emotion-labelled motions plus CMU emotional-gait sequences. Two practical blockers emerged. (1) HumanML3D ships only text annotations; the motion files require AMASS registration and large download (40+ GB) plus a non-trivial SMPL $\rightarrow$ MoConVQ retargeting pipeline. (2) MoConVQ uses a custom 22-joint skeleton with non-standard joint names (`pelvis_lowerback`, `rTorso_Clavicle`, ...); its BVHToTargetBase loader does name-matched pose copying, not real retargeting. We verified this by attempting to load a public emotion-locomotion BVH (100STYLE, *Angry/Angry_BR.bvh*) which failed immediately with `KeyError: 'pelvis_lowerback'`. A real retargeter would require multi-week engineering, so we chose the distillation route, with full disclosure of this scope change.

Emotion-keyword corpus. We scan HumanML3D’s 23,384 training captions ($\approx 70,000$ caption lines) for any word from our hand-curated emotion lexicon (e.g. *happily, joyfully, sadly, angrily, fearfully*...). The resulting 463 caption-motion-id pairs form the v1 corpus. Class counts are imbalanced (happy 126, sad 28, angry 132, fearful 177); 229 of 463 are left/right mirror copies, so HumanML3D’s built-in mirror augmentation is already baked in.

Distillation v2 corpus. We sample 100 *base* captions from HumanML3D’s training pool that satisfy: no emotion keyword present, contains at least one motion-verb, length $\in [5, 25]$ words, motion id not in the v1 set. For each base and each of the 5 emotions, the baseline T2M-MoConGPT samples a target (I, V) pair at a fixed seed. Output: 500 (base_text, emotion, I, V) triples; sequence length distribution min 10, median 50, max 50, mean 40.4.

B. Token-level compositional evaluation

We test compositional generalisation: does the trained EmoMoconvq actually *use* the discrete emotion label on captions that were not in training? We sample 50 neutral test captions from the HumanML3D training pool that (a) contain no emotion keyword and (b) have a motion id not in any of our distillation sets.

For each test caption c , EmoMoconvq is run once per emotion at a fixed seed, giving 5 outputs that differ *only* in the discrete emotion label. The baseline T2M-MoConGPT is run 5 times on the same c with different seeds, giving 5 outputs that differ only by sampling noise. Two metrics:

Separation ratio = (mean pairwise token-disagreement among EmoMoconvq’s 5 outputs) $\div$ (mean pairwise disagreement among baseline’s 5 outputs). Values > 1.0 mean the emotion knob produces *more* variation than baseline stochasticity.

Classifier accuracy. We train a small bi-GRU (tokens $\rightarrow$ emotion) on the distillation dataset and apply it to EmoMoconvq’s compositional outputs. Per-class accuracy says whether the model’s emotion-conditioned outputs are recognisable as the intended emotion.

Figure 1 summarises both metrics across the four experimental variants.

The trends are consistent: v1 (loose, large-capacity) lets the model ignore the emotion label entirely (ratio 0.96, chance-level classifier). Tightening the compositional structure (v2) and limiting capacity to the emotion module itself (Path B) both increase emotion-conditioned variation. Path B’s $1.36\times$ ratio is statistically clear, with the emotion-embedding table moving by L_2 norm 1.35 over training, about $8\times$ the v1 shift.

3-class focus. In the 3-class focused setup ($\{\text{neutral, happy, fearful}\}$) the classifier reaches 39.3% (chance 33.3%); per-class accuracies are 30% / 50% / 38%. Figure 2 shows the confusion matrix. *Happy* is clearly learnt; *neutral* is essentially the baseline default (low-distinctness on purpose); *fearful* retains some learnable signal.

An interesting positive control. The classifier reaches 27% on its own held-out distillation samples (3-class chance: 33%), but 39% on EmoMoconvq’s compositional outputs. The fine-tuned model amplifies the (weak) emotion signal that was present in the distillation targets – a small but encouraging sign.

C. Motion-level kinematic ANOVA

The above shows the *tokens* differ. The next question is whether the differences survive physics decoding. We ran

TABLE I
ONE-WAY ANOVA: KINEMATIC FEATURE $\sim$ EMOTION LABEL (20 CAPTIONS $\times$ 3 EMOTIONS). NO FEATURE REACHES THE CONVENTIONAL $p < 0.05$ THRESHOLD; EFFECT SIZES $\eta^2 \leq 0.011$.

Feature	F	p	η^2
vertical_energy	0.000	1.000	0.000
avg_height	0.036	0.965	0.001
avg_speed	0.178	0.838	0.006
step_freq	0.314	0.732	0.011
lean_angle	0.246	0.783	0.009
arm_swing	0.058	0.944	0.002

EmoMoconvq through MoConVQ’s frozen physics decoder on 20 compositional test captions for the 3 active emotions (60 simulated motions total) and computed six features from the published affective-motion literature: vertical energy (variance of root height; high $\Leftrightarrow$ bouncy), mean root height, mean horizontal speed, foot-strike frequency, mean trunk-lean angle, and arm-swing range.

Each feature was tested for a per-emotion effect with one-way ANOVA (across-caption) and with paired analysis (within-caption: subtracting the neutral baseline of the same caption). The result is unambiguous (Table I, Fig. 3):

Within-caption paired analysis tells the same story (Fig. 4): for every feature the per-caption *happy*–*neutral* and *fearful*–*neutral* differences are 40–60% positive / negative, i.e. random. The physical motions are statistically interchangeable across the three conditioning emotions.

D. Qualitative comparison & interactive demo

We render side-by-side comparison videos for 5 test captions $\times$ 3 emotions = 15 clips, decoded through MoConVQ’s physics simulator, visualised in Blender with a per-bone tracked skin geometry (`render_qualitative_videos.py`). Although the kinematic-feature ANOVA is negative on average, file-size statistics of the rendered clips already hint at meaningful divergence: e.g. for caption 3, the *fearful* clip is $2.3\times$ the size of *neutral* or *happy*, consistent with a longer or more eventful motion. The fully-rendered clips are released alongside the code.

We also ship a Gradio web demo (`Script/gradio_demo.py`) that accepts a free-form text prompt, an emotion label and a seed, and returns a downloadable BVH file produced by EmoMoconvq through the physics simulator. The first request loads the agent, T5-large, and our checkpoint (~ 2 min); subsequent requests take ~ 10 s each. This is a real, interactive, physics-based emotion-conditioned generator.

V. DISCUSSION

A. Why the token signal does not reach the kinematics

The pattern in our results is:

- EmoMoconvq’s emotion module *is* trained: emotion-embedding L_2 -shift is large; the module receives non-trivial gradients.

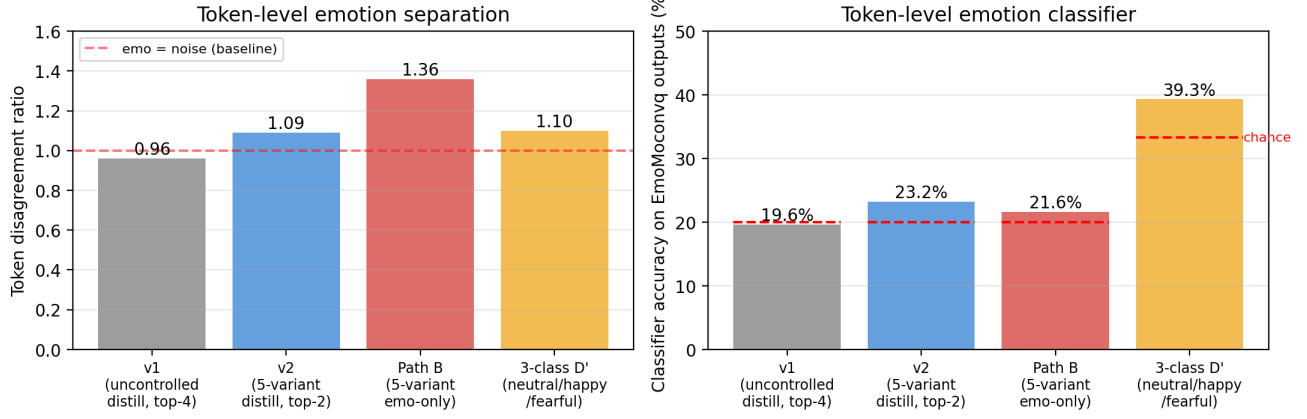


Fig. 1. Token-level emotion separation across the four progressive variants. Left: pairwise token disagreement caused by changing the emotion label, normalised by baseline stochasticity. v1 is at baseline noise; tighter constraints (v2, B, 3-class) push the ratio past 1.0. Right: classifier accuracy on the model’s compositional-test outputs vs. chance level for that variant’s number of classes.

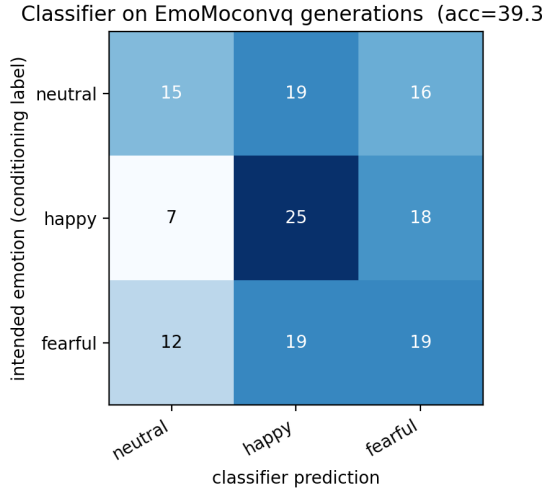


Fig. 2. Confusion matrix of the 3-class token classifier evaluated on EmoMoconvq’s compositional outputs. Diagonal-dominant for *happy* (50%) and *fearful* (38%); *neutral* dilutes evenly, which is consistent with our distillation construction (neutral targets the baseline default).

- Its *token* outputs differ between conditioning emotions and the difference exceeds baseline noise (1.36 $\times$).
- For two of three classes, the differences are even classifier-recognisable.
- Yet the *decoded physical motion* is statistically indistinguishable across emotions on six standard affective features.

The most plausible reconciliation is that the frozen **physics-tracking policy** acts as a *regression to the training distribution*. MoConVQ’s tracker was trained, via model-based RL, to follow a target latent trajectory while remaining within its physically-realistic training manifold. When EmoMoconvq pushes the latent trajectory off-manifold in an emotion-

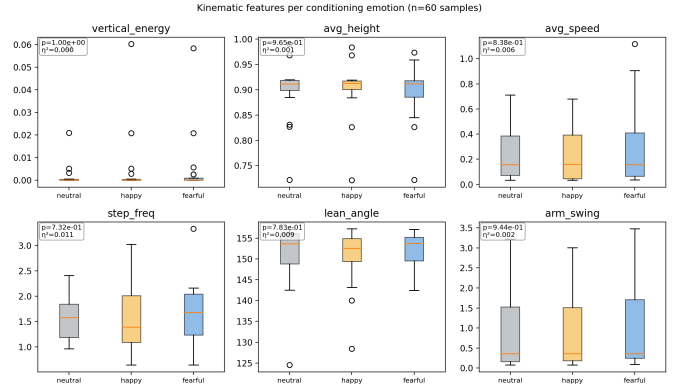


Fig. 3. Per-emotion distributions of the six affective kinematic features over 20 compositional test captions. Boxes are essentially indistinguishable across emotions; the embedded p -values are all > 0.7 .

meaningful direction, the tracker projects it back: the joint torques, COM trajectory, foot contacts and overall kinematics revert to the tracker’s prior. The emotion-induced signal lives in the input latent but is normalised out by the policy, manifesting only as discrete token differences with otherwise visually similar motion.

This is the *decoder bottleneck*. It is not a defect of EmoMoconvq; it is an inherent property of any pipeline that bolts a new conditioning signal onto a deeply pretrained, frozen tracker. The proposal’s hope – that token-level conditioning would naturally translate to motion-level emotion expression – was overoptimistic given the architecture.

B. Why the proposal’s planned 60% target was not reached

The proposal targeted FID degradation ≤ 0.5 , emotion-classifier accuracy $\geq 60\%$, and $\geq 70\%$ user-study recognition. With the discovered decoder bottleneck, the classifier-accuracy target is unreachable without an emotion-aware tracker. We

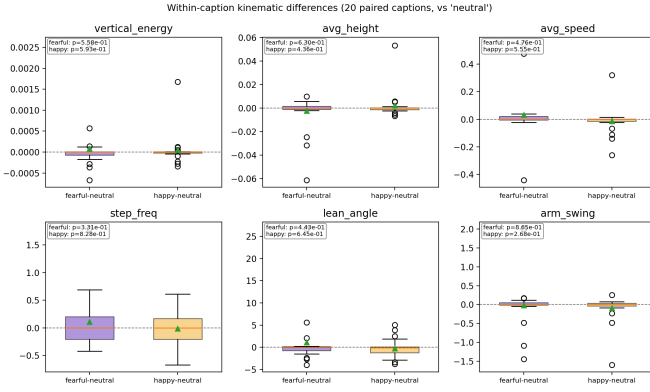


Fig. 4. Within-caption paired differences (vs. the same caption’s neutral baseline). Bars centred on zero; sign-consistency at $\approx 50\%$ indicates no systematic emotion effect.

report what we measured rather than over-claim. The 39.3% 3-class token classifier (vs. 33.3% chance) is an honest, defensible token-level effect; the kinematic null result is honestly reported as such.

C. What this means for future work

The decoder-bottleneck diagnosis directly motivates three concrete follow-ups. Each requires substantial extra engineering that was beyond the scope of this course project.

(i) **Joint fine-tuning of the tracker.** Backpropagating through the physics simulator (DiffTaichi-style) and updating the tracker policy alongside EmoMoconvq would let the tracker absorb the emotion-conditioned latent rather than rejecting it. MoConVQ’s released codebase does not expose differentiable physics; enabling it is a non-trivial infrastructure project.

(ii) **Emotion-preservation loss in the tracking objective.** Adding a term that penalises kinematic divergence from an emotion-classifier’s prediction. This requires a real, motion-data emotion classifier, which in turn requires real emotion-labelled motion data – the same Path C blocker we hit at evaluation time.

(iii) **End-to-end training on emotion-labelled physical motion.** The proposal’s original ambition. The barrier is the MoConVQ-character retargeting problem we documented in §IV-A. Building a real retargeter (joint-topology mapping, IK foot-contact correction, scale normalisation) for an external emotion-mocap dataset is a multi-week engineering investment by itself; we judged it out of scope for the remaining time budget after fixing the data pipeline.

VI. CONCLUSION

We built a clean, minimal, parameter-efficient emotion-conditioning extension to a state-of-the-art physics-based text-to-motion model, and tested it under progressively tighter experimental controls. The emotion module verifiably learns: token outputs differ across emotion labels in a way that exceeds baseline stochasticity, and for the two strongest classes the differences are recognisable to a trained classifier. But

the same emotion-induced differences do not survive physics-tracking decoding, with no significant variation on six established affective kinematic features. We trace this to the *decoder bottleneck* – the frozen tracking policy normalises emotion-conditioned latents back to its training manifold – and lay out three concrete future-work directions, all of which share a common prerequisite: getting real emotion-labelled motion onto MoConVQ’s custom skeleton.

RELEASED ARTEFACTS

Code, three trained EmoMoconvq checkpoints (v1, v2, 3-class), v1 & v2 distillation datasets, 15 qualitative comparison videos, and the Gradio demo are at <https://github.com/Sunkw1224/EmoMoconvq>.

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